Supplementary Materials

Section A. Data processing and variable definitions

Protected area data

Protected area boundaries and attributes are drawn from the World Database on Protected Areas (WDPA), May 2021 release (UNEP-WCMC and IUCN 2023). The following WDPA fields are used, with definitions drawn from the WDPA User Manual (UNEP-WCMC 2024):

- **STATUS**: The legal or established status of a protected area. We retain only sites with `STATUS` equal to "Designated", "Established", or "Inscribed", which correspond to sites that have been formally recognized through legal or other effective means.
- **STATUS_YR**: The year in which the protected area was designated or established. A value of **0 indicates that the designation year was not reported** by the data provider. It does not mean the PA was created in year zero, nor that its existence is uncertain – it simply reflects missing temporal metadata. In the May 2021 WDPA release, a non-negligible number of PAs carry `STATUS_YR = 0`.
- **IUCN_CAT**: The IUCN management category assigned to the protected area. Categories Ia, Ib, II, and III correspond to strict protection regimes (nature reserves, wilderness areas, national parks). Categories IV, V, and VI correspond to less restrictive management (habitat management areas, protected landscapes, sustainable-use areas). Many PAs in the WDPA carry a category of "Not Reported", "Not Applicable", or "Not Assigned" – we group these as "unknown IUCN category".
- **DESIG_ENG**: The designation type in English. UNESCO-MAB Biosphere Reserves are excluded, as is standard practice, because they often encompass large areas with minimal legal protection (Hanson 2022).
- **MARINE**: Indicates whether the PA is terrestrial, coastal, or marine. We exclude purely marine PAs (`MARINE = "2"`).

Temporal classification of protected areas

Because `STATUS_YR = 0` is common in the WDPA, we distinguish two sets of PAs:

1. **PAs with recorded designation year**: used for the 2000–2020 comparison. For 2000, we retain PAs with `1 ≤ STATUS_YR ≤ 2000`. For 2020, we retain PAs with `1 ≤ STATUS_YR ≤ 2020`.
2. **All PAs in 2020**: includes all PAs from set (1) above plus those with `STATUS_YR = 0`. These PAs are known to exist because they appear in the May 2021 WDPA release – their designation year is simply not recorded. This is the set used for the main 2020 cross-section (Table 1, Figure 1).

The 2000–2020 comparison is restricted to set (1) because PAs with missing designation year cannot be reliably assigned to the 2000 period.

Spatial processing in Google Earth Engine

Population counts inside and near protected areas were computed in Google Earth Engine (GEE) for each country and each first-level administrative unit (ADM1) from geoBoundaries v6.0.0 (Runfola et al. 2020). For each administrative unit, the following steps were performed:

1. **Classification**: PAs were assigned to one of three IUCN groups (strict, non-strict, unknown). Within each group, a binary raster mask was constructed at the resolution of the population grid.
2. **Hierarchical assignment**: Where PA categories overlap spatially, pixels were assigned to the highest-priority category using an exclusive hierarchy: strict > non-strict > unknown. Each pixel belongs to at most one category.
3. **Buffer construction**: For each category mask, a 10 km buffer was computed using `focal_max`. The buffer represents the ring of pixels within 10 km of the PA boundary but not inside any PA. Buffer pixels are also assigned exclusively – a pixel already claimed by a higher-priority PA or buffer is not counted again.
4. **Zonal aggregation**: Population (from GHSL or WorldPop) and land area were summed within each mask and buffer, clipped to national boundaries. Buffer zones are truncated at international borders.

Output dataset structure

The GEE computation produces one CSV file per country and population source (GHSL or WorldPop), with one row per ADM1 unit and temporal subset. Key variables in the output:

Variable	Definition	Unit
`pop_total`	Total population within the administrative unit	persons
`pop_strict`	Population inside strict PAs (IUCN Ia-III)	persons
`pop_nonstrict`	Population inside non-strict PAs (IUCN IV-VI)	persons
`pop_unknowncat`	Population inside PAs of unknown IUCN category	persons
`pop_strict10`	Population in the 10 km buffer ring around strict PAs (exclusive of inside)	persons
`pop_nonstrict10`	Population in the 10 km buffer ring around non-strict PAs (exclusive)	persons
`pop_unknowncat10`	Population in the 10 km buffer ring around unknown-category PAs (exclusive)	persons

Variable	Definition	Unit
`area_strict`	Land area inside strict PAs	km²
`area_nonstrict`	Land area inside non-strict PAs	km²
`area_unknowncat`	Land area inside unknown-category PAs	km²
`scenario`	Temporal subset: `Confirmed_2000`, `Confirmed_2020`, or `Unknown_Year`	–
`source`	Population dataset: `GHSL` or `WP` (WorldPop)	–

A separate national-level aggregation file provides total PA area per country (all categories combined, no temporal filter).

In the R analysis, "inside or within 10 km" is computed as the sum of the inside population and the buffer-ring population (e.g., `pop_strict + pop_strict10`). Percentage shares are computed relative to `pop_total`, which represents the total national population from the same gridded dataset.

References

Hanson, Jeffrey O. 2022. "Wdpar: Interface to the World Database on Protected Areas." *Journal of Open Source Software* 7 (78): 4594. https://doi.org/10.21105/joss.04594.

Runfola, Daniel, Austin Anderson, Heather Baier, Matt Crittenden, Elizabeth Dowker, Sydney Fuhrig, Seth Goodman, et al. 2020. "geoBoundaries: A Global Database of Political Administrative Boundaries." *PLOS ONE* 15 (4): e0231866. https://doi.org/10.1371/journal.pone.0231866.

UNEP-WCMC. 2024. "World Database on Protected Areas (WDPA) [WDOECM] User Manual 1.6." Cambridge, United Kingdom: UNEP World Conservation Monitoring Centre. https://wdpa.s3.amazonaws.com/WDPA_Manual/English/WDPA_WDOECM_Manual_1_6.pdf.

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Section B. Country-level detail

Table S1: Protected area coverage and population proximity (2000–2020)

Confirmed PAs only, GHSL estimates

	PA area (km^2)		% pop. inside PAs		% pop. inside or within 10 km	
Country	2000	2020	2000	2020	2000	2020
Afghanistan	293	21,490	0.1	1.1	0.9	14.9
Angola	82,181	82,234	1.5	1.7	2.9	4.0
Bangladesh	3,862	3,401	0.4	0.5	4.9	11.1
Benin	24,280	24,281	1.6	1.7	23.6	23.7
Bhutan	12,420	16,493	3.4	3.5	55.2	56.2
Bolivia	205,366	225,472	7.1	13.4	51.8	56.7
Burkina Faso	36,010	39,287	3.1	3.7	31.1	42.8
Burundi	1,167	1,169	0.1	0.1	31.8	34.9
Cabo Verde	0	19	0.0	0.1	0.0	42.1
Cambodia	23,880	57,709	1.7	4.1	13.4	30.8
Cameroon	23,875	47,007	0.1	2.0	4.5	25.2
Central African Republic	66,175	66,198	0.2	0.3	19.2	20.8
Chad	141,013	170,925	3.7	3.6	14.3	21.8
Comoros	12	522	0.5	11.2	35.7	99.5
Congo, Dem. Rep.	160,176	280,598	1.4	4.7	22.5	26.4
Congo, Rep.	31,154	117,199	0.5	12.0	49.1	73.9
Côte d'Ivoire	20,912	21,197	0.1	0.3	25.8	29.0
Djibouti	0	200	0.0	0.1	0.0	3.3
Egypt, Arab Rep.	41,209	108,790	0.0	0.2	12.6	13.2
El Salvador	15	1,241	0.0	2.3	4.6	90.0
Eritrea	5,810	5,851	1.1	1.4	7.9	6.5
Eswatini	694	715	0.2	0.1	35.7	42.9
Ethiopia	101,516	163,515	2.6	6.3	7.6	13.8
Gambia, The	252	371	0.0	0.3	55.4	78.1
Ghana	27,604	27,979	0.8	0.8	69.7	70.2
Guinea	10,771	53,781	1.9	14.1	44.8	56.2
Guinea-Bissau	3,247	5,503	1.8	4.9	12.6	17.6
Haiti	548	1,677	0.9	2.7	11.0	50.8

Source: WDPA May 2021, GHSL.

Table S1: Protected area coverage and population proximity (2000–2020)

Confirmed PAs only, GHSL estimates

	PA area (km^2)		% pop. inside PAs		% pop. inside or within 10 km	
Country	2000	2020	2000	2020	2000	2020
Honduras	21,326	23,397	3.1	4.1	69.6	78.1
Indonesia	40,691	63,128	0.1	0.3	3.2	5.5
Kenya	53,815	54,836	2.8	3.1	52.7	53.6
Kiribati	0	0	0.0	0.0	0.0	0.0
Korea, Dem. Rep.	0	47	0.0	0.0	0.1	1.0
Kyrgyz Republic	4,550	4,756	0.4	0.4	16.9	16.6
Lao PDR	23,384	39,313	1.5	3.1	27.9	39.0
Lesotho	80	80	0.0	0.0	11.6	12.0
Liberia	1,559	2,750	0.0	0.8	3.7	10.0
Madagascar	30,101	42,399	0.9	3.8	9.7	27.4
Malawi	16,347	16,882	1.3	1.4	48.5	47.4
Mali	41,151	81,064	3.2	8.9	36.9	50.9
Mauritania	5,543	5,543	0.3	0.1	1.0	0.5
Micronesia, Fed. Sts.	0	0	0.0	0.0	0.0	0.0
Moldova	10	10	0.0	0.0	8.6	13.3
Mongolia	218,574	286,296	1.6	1.9	36.2	54.4
Morocco	11,129	16,899	0.5	1.2	34.1	42.1
Mozambique	74,188	135,976	0.8	4.9	5.8	25.3
Myanmar	10,529	43,249	0.2	0.4	9.4	13.9
Nepal	23,679	28,392	0.7	2.6	24.5	36.0
Nicaragua	15,378	22,932	2.2	4.0	65.7	71.4
Niger	79,380	191,551	1.8	4.3	4.1	9.4
Nigeria	32,895	40,763	0.2	0.5	2.6	4.6
Pakistan	81,000	81,297	2.9	2.8	23.3	23.4
Palestine (West Bank and Gaza)	1	1	0.0	0.0	66.7	75.2
Papua New Guinea	12,925	16,646	0.9	1.0	10.5	11.4
Philippines	21,699	37,365	3.0	3.0	39.5	56.6
Rwanda	1,596	1,697	0.0	0.1	19.6	26.2
Senegal	21,801	23,063	1.2	1.1	16.0	34.8

Source: WDPA May 2021, GHSL.

Table S1: Protected area coverage and population proximity (2000–2020)

Confirmed PAs only, GHSL estimates

	PA area (km^2)		% pop. inside PAs		% pop. inside or within 10 km	
Country	2000	2020	2000	2020	2000	2020
Sierra Leone	2,558	3,470	0.6	1.0	26.4	32.1
Solomon Islands	114	156	0.3	0.3	0.4	1.6
Somalia	9	62	0.0	0.0	0.2	0.1
Sudan	26,239	26,810	0.1	0.1	0.3	0.3
Syrian Arab Republic	2	43	0.0	0.0	0.1	0.8
São Tomé and Príncipe	0	312	0.0	0.0	0.0	47.6
Tajikistan	25,203	25,675	0.6	0.7	13.4	15.7
Tanzania	188,495	219,550	2.0	3.0	42.4	43.7
Togo	6,205	6,176	2.5	2.7	51.9	67.1
Tunisia	7,453	11,483	0.1	0.2	31.1	62.2
Uganda	26,786	29,261	1.7	2.1	85.3	86.2
Ukraine	15,251	17,580	0.1	0.0	51.5	53.6
Uzbekistan	12,364	19,319	0.1	0.3	3.7	8.7
Vanuatu	53	53	0.1	0.1	23.0	26.1
Viet Nam	5,565	20,310	0.5	0.9	10.4	22.5
Yemen, Rep.	0	2,765	0.0	0.0	0.0	0.2
Zambia	231,586	235,429	8.0	8.0	73.0	72.4
Zimbabwe	46,886	74,458	0.6	3.9	14.9	34.0

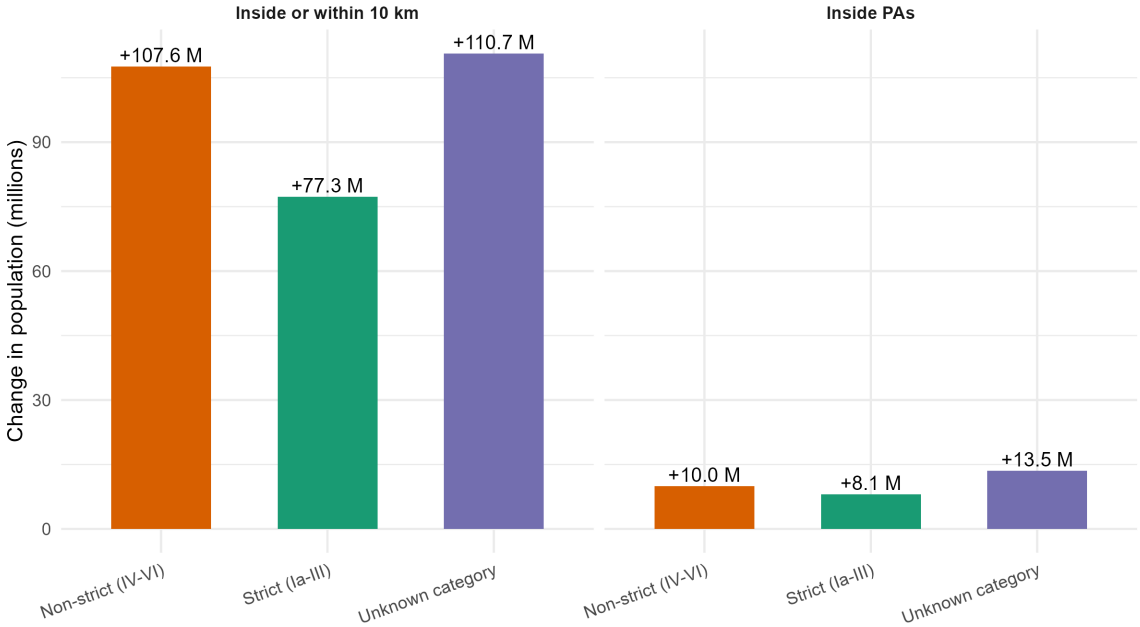
Source: WDPA May 2021, GHSL.

Section C. Net change in population near protected areas, 2000–2020

Figure S1 shows the net change (2020 minus 2000) in population inside protected areas and within 10 km of their boundaries, for the 75 LMICs combined, decomposed by IUCN management category. This figure is restricted to protected areas with recorded designation years in the WDPA.

Figure S1: Net change in PA-adjacent population (2000-2020)

75 LMICs (excl. India), GHSL. PAs with recorded designation year only.



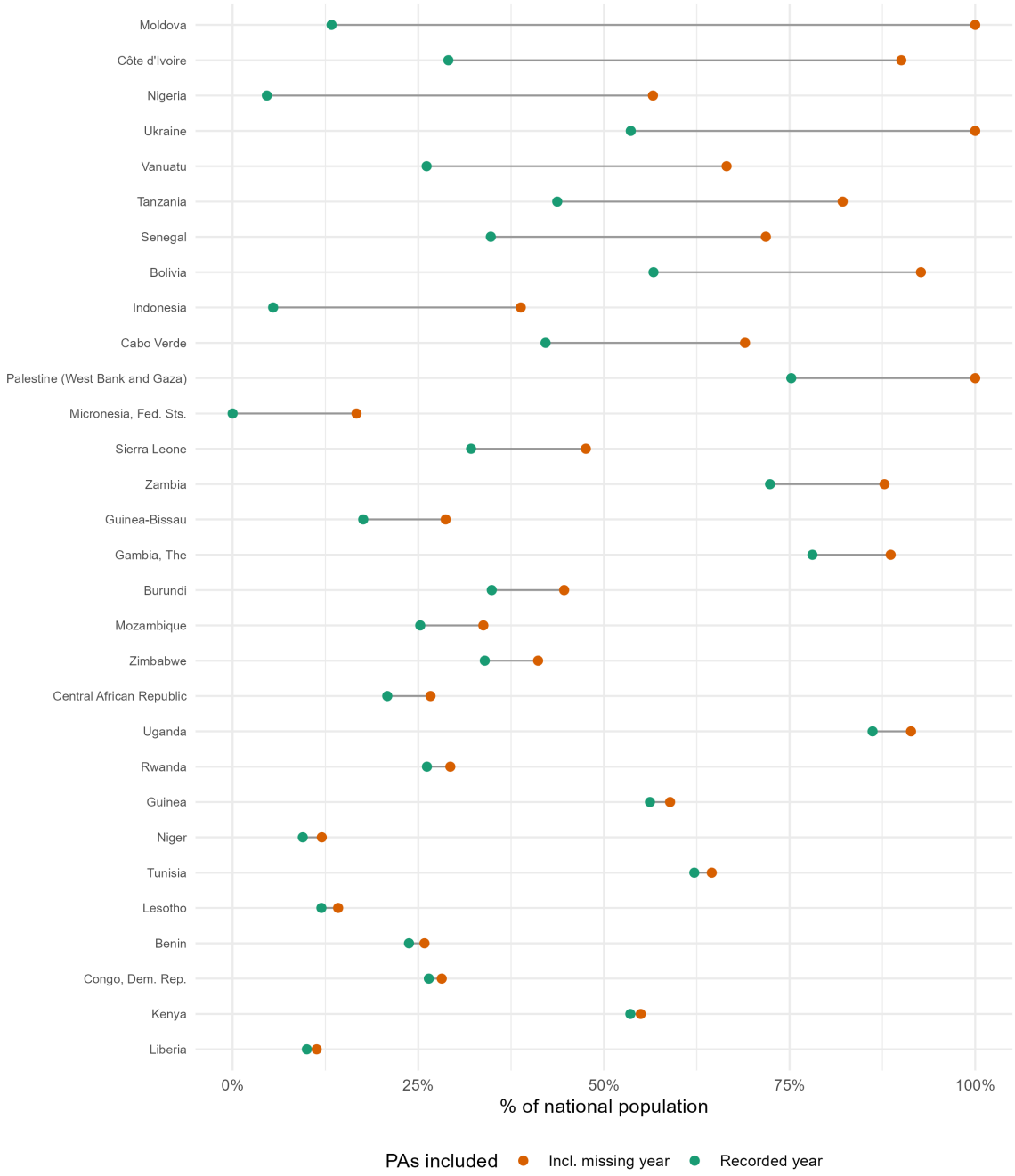
Section D. Effect of including PAs with missing designation year

Figure S2 displays, for each country, the share of the national population residing inside PAs or within 10 km under two counts: one restricted to PAs whose designation year is recorded as falling on or before 2020 ("Recorded year"), and one that also includes PAs with missing designation year ("Incl. missing year"). The gap between the two bars represents the uncertainty attributable to incomplete temporal metadata in the WDPA.

Figure S2: Effect of including PAs with missing designation year

% of national population inside PAs or within 10 km (GHSL 2020)

Top 30 countries by gap



Section E. Largest GHSL-WorldPop discrepancies

Table S2: Largest Differences Between GHSL and WorldPop Estimates

All PAs in 2020 (incl. unknown designation year). Absolute difference > 5 pp and relative difference > 10%

Country	Perimeter	GHSL (%)	WorldPop (%)	Abs. diff. (pp)	Rel. diff.
Comoros	Inside PAs	11.2	37.7	26.4	235%
Moldova	Inside or within 10 km	102.2	88.9	13.2	13%
Congo, Rep.	Inside PAs	12.0	24.1	12.0	100%
São Tomé and Príncipe	Inside or within 10 km	47.6	58.1	10.5	22%
Djibouti	Inside or within 10 km	3.3	12.6	9.3	279%
São Tomé and Príncipe	Inside PAs	0.0	8.8	8.8	18,566%
Guinea-Bissau	Inside or within 10 km	28.7	35.1	6.4	22%
Bhutan	Inside PAs	3.5	9.0	5.5	157%
Nepal	Inside or within 10 km	36.0	41.4	5.4	15%
Zambia	Inside PAs	8.6	13.9	5.3	62%
Senegal	Inside PAs	2.6	7.7	5.0	192%

Source: WDPA May 2021, GHSL & WorldPop 2020.